EVERY $m \times 3$ BIQUADRATIC SUM OF SQUARES IS A SUM OF $2m$ SQUARES

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ABSTRACT. Let $\mathrm{BSR}(m,n)$ be the largest number of squares of real bilinear forms needed to write an $m \times n$ biquadratic sum of squares. We prove $\mathrm{BSR}(m,3) = 2m$ for every $m \geq 2$, answering the question of Qi, Löfberg and Chen on the exact values of $\mathrm{BSR}(m,3)$; in particular $\mathrm{BSR}(4,3) = 8$, which closes the interval $8 \leq \mathrm{BSR}(4,3) \leq 9$. The cases $m = 2,3$ already follow from results of Blekherman, Plaumann, Sinn, Smith, Velasco and Vinzant; the new range is $m \geq 4$. The lower bound comes from the quadratic persistence $\mathrm{qp}(\mathbb{P}^{m-1} \times \mathbb{P}^2) = m$ and an explicit form. The upper bound follows the boundary strategy of Blekherman, Dunbar and Sinn for ternary forms: at a hypothetical boundary point of the set of sums of $2m$ squares inside the cone, a Hankel multiplier satisfies first and second order conditions. We replace their classification of Gorenstein ideals by a rank inequality for such multipliers and an induction on m, closed by a second variation built from the alternating symbol in three indices.

Companion site. https://karlesmarin.github.io/blueprints/bsr-width-three/: every statement with its dependencies and its checks, and labs in the browser. This note: doi:10.5281/zenodo.24048317 (Zenodo), with its supplementary material.

1. The result

Write $x = (x_1, \ldots, x_m)$, $y = (y_1, y_2, y_3)$. A *biquadratic form* is a real form of bidegree $(2,2)$ in (x,y), and $\mathrm{BSR}(m,n)$ is the least r such that every biquadratic sum of squares in $m+n$ variables is a sum of r squares of bilinear forms. It is the Pythagoras number of the coordinate ring of the Segre variety $X = \mathbb{P}^{m-1} \times \mathbb{P}^{n-1}$ in degree two [4]. By [4, Theorem 1.3], $\mathrm{BSR}(m,n) \geq mn - \mathrm{qp}(X)$; for $n = 3$ the persistence is $\mathrm{qp}(X) = m$ [6, Theorem 1], so $\mathrm{BSR}(m,3) \geq 2m$. The parallel-line covering bound of [4, Corollary 2.15] gives $\mathrm{BSR}(m,n) \leq mn - \max(m,n) + 1$ (see [6, Theorem 9]), hence $2m \leq \mathrm{BSR}(m,3) \leq 2m+1$ for $m \geq 3$; the theorem below removes the last unit. For $m = 4$ the lower bound 8 was first given by an explicit form of Xu, Cui and Qi [8, Theorem 4.1], and Qi, Löfberg and Chen ask for the exact values of $\mathrm{BSR}(m,3)$ [7, §9].

Theorem 1. *For every $m \geq 2$, $\mathrm{BSR}(m,3) = 2m$.*

For $m = 2$ the Segre variety $\mathbb{P}^1 \times \mathbb{P}^2$ has minimal degree and $\mathrm{BSR}(2,3) = 4$ by [3] and [4, Theorem 1.4]; for $m = 3$, $\mathbb{P}^2 \times \mathbb{P}^2$ is arithmetically Cohen–Macaulay of degree $6 = \mathrm{codim} + 2$ and $\mathrm{BSR}(3,3) = 6$ by [4, Theorem 1.5]. The statement is new for $m \geq 4$. The lower bound is §6; the rest of the note proves the upper bound.

Strategy and attribution. The upper bound follows the boundary strategy of Blekherman, Dunbar and Sinn [2, §§2–3] (see also [3, 5]): if not every sum of squares is a sum of $2m$ squares, a generic point of the boundary of the set of sums of $2m$ squares inside the interior of the cone carries a nonzero linear functional (a Hankel multiplier) satisfying first and second order

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optimality conditions. In [2] the contradiction comes from Diesel's classification of height-three Gorenstein ideals; here it comes from a rank inequality for the multiplier (Lemma 5) and an induction on m whose last step is a second variation built from the alternating symbol (§5).

2. Setting

Let $A = \mathbb{R}^m$, $B = \mathbb{R}^3$ and $V = A \otimes B$, identified with the space of bilinear forms $h(x, y)$ through the monomial basis and the Euclidean inner product; $x \otimes y \in V$ is a simple tensor. Let R_2 be the space of biquadratic forms, of dimension $N = \binom{m+1}{2} \cdot 6 = 3m(m+1)$. For $k \geq 1$ put

$$\Phi_k \colon V^k \to R_2, \qquad \Phi_k(h_1, \dots, h_k) = h_1^2 + \cdots + h_k^2,$$

$\Sigma_k = \Phi_k(V^k)$ and $\Sigma = \bigcup_k \Sigma_k$, the cone of biquadratic sums of squares.

(1) $\Sigma = \Sigma_{3m}$: a form in Σ has a positive semidefinite Gram matrix of size $3m = \dim V$, hence of rank at most $3m$.
(2) Σ_k is closed: the sum of the coefficients of $x_i^2 y_j^2$ in $\Phi_k(h)$ is the sum of the squares of all coefficients of all h_i, so preimages of a bounded set are bounded.
(3) Σ is a closed convex cone with nonempty interior, since squares of bilinear forms span R_2 by polarization. A form in the interior is strictly positive at every real point of X, because evaluation at a real point is a nonzero linear functional that is nonnegative on Σ.

To $\lambda \in R_2^*$ associate the symmetric bilinear form $H_\lambda(u, v) = \lambda(uv)$ on V; $\lambda \mapsto H_\lambda$ is injective, and $H = H_\lambda$ has the *Hankel symmetry*

$$H(x \otimes b,\, x' \otimes c) = H(x \otimes c,\, x' \otimes b) \qquad (x, x' \in A,\ b, c \in B), \tag{1}$$

since both sides are λ of the same biquadratic form. In the basis of B, (1) says that the 3×3 array of $m \times m$ blocks H_{jl} of H has symmetric blocks and $H_{jl} = H_{lj}$. The differential of Φ_k is $d\Phi_k(h)(D) = 2\sum_i h_i D_i$, so

$$\lambda \circ d\Phi_k(h) = 0 \iff W_h := \langle h_1, \dots, h_k \rangle \subseteq \ker H_\lambda. \tag{2}$$

3. A regular point and a boundary point

Lemma 2. *For every $m \geq 2$ there are $h \in V^{2m}$ with integer coefficients at which $d\Phi_{2m}$ is onto. In particular Σ_{2m} has nonempty interior.*

Proof. Identify V with A^3 by the three columns (y_1, y_2, y_3). Let P be the cyclic permutation matrix of size m, $D = \operatorname{diag}(1, \dots, m)$, $Q = DP^{-1}$ and $C = [\, I \ \ P \ \ Q \,] \colon A^3 \to A$. Take for h the integer basis $(-Pe_i, e_i, 0)$, $(-Qe_i, 0, e_i)$, $1 \leq i \leq m$, of $\ker C$, so $W_h = \ker C$. By (2) we must show that a symmetric H with (1) and $\ker C \subseteq \ker H$ vanishes. Such an H is $C^\top M C$ with M symmetric, with blocks M, MP, MQ, $P^\top MP$, $P^\top MQ$, $Q^\top MQ$. Symmetry of the off-diagonal blocks gives $MP = P^\top M$, $MQ = Q^\top M$ and $P^\top MQ = Q^\top MP$. Then $P^\top MQ = P^\top Q^\top M$ and $Q^\top MP = Q^\top P^\top M$, so $(PQ - QP)^\top M = 0$, that is $M(PQ - QP) = 0$. Now $PQ - QP = PDP^{-1} - D$ is diagonal with entries $d_{\sigma(i)} - d_i = \pm 1$ or $\pm(m-1)$, hence invertible, and $M = 0$. □

Proposition 3. *Suppose $\Sigma_{2m} \neq \Sigma$. Then there are $F \in \Sigma_{2m}$ in the interior of Σ and $\lambda \in R_2^* \setminus 0$, $H = H_\lambda$, such that for every $h \in V^{2m}$ with $\Phi_{2m}(h) = F$:*

(1) $W_h \subseteq \ker H$ *and* $\lambda(F) = 0$*;*
(2) $\sum_i H(D_i, D_i) \geq 0$ *for every* $D \in V^{2m}$ *with* $\sum_i h_i D_i = 0$*;*
(3) H *is indefinite;*
(4) *the* h_i *have no common zero* $([x], [y]) \in X(\mathbb{C})$.

Proof. This is the argument of [2, Proposition 2.2 and Proposition 3.3]; we give it in our setting. Let Ω be the interior of Σ. The set $\Omega \cap \Sigma_{2m}$ contains a nonempty open set (Lemma 2 and the submersion theorem), and $\Omega \setminus \Sigma_{2m}$ is open and nonempty. Fix a semialgebraic stratification of R_2 compatible with Σ_{2m} and Ω [1, §9]. A generic segment in the convex set Ω joining the two open sets avoids every stratum of dimension at most $N-2$ and meets the strata of dimension $N-1$ transversally; where it leaves Σ_{2m} it crosses a stratum $S \subseteq \partial\Sigma_{2m}$ of dimension $N-1$ with Σ_{2m} on one side and its complement on the other. Near a point of S there is a smooth ρ with $\rho = 0$ on S, $d\rho \neq 0$ and $\rho \geq 0$ on Σ_{2m}.

We choose $F \in S$ outside the set Z of forms in Ω that are singular at a point of $X(\mathbb{C})$. A form in Ω is positive on $X(\mathbb{R})$, and if one factor of a singular point $p = ([x], [y])$ were real the other could be taken real by linearity, so p and $\bar{p}$ differ in both factors. After complex changes of coordinates $p, \bar{p}$ become $(e_1, f_1), (e_2, f_2)$; singularity at (e_i, f_i) fixes the coefficients of $x_i^2 y_i^2$, $x_i x_j y_i^2$ $(j \neq i)$ and $x_i^2 y_i y_l$ $(l \neq i)$, which are $m+2$ conditions, and the two lists are disjoint. For real forms this is $2(m+2)$ real conditions, while p moves in $2(m+1)$ real dimensions; hence $\dim Z \leq N-2 < \dim S$.

Put $\lambda = d\rho_F$. For a preimage h, $\rho \circ \Phi_{2m} \geq 0$ near h and vanishes at h, so it has a local minimum there: the first order condition is $\lambda \circ d\Phi_{2m}(h) = 0$, which is (1) by (2), and $\lambda(F) = \sum_i H(h_i, h_i) = 0$. For D in the kernel of $d\Phi_{2m}(h)$ the second derivative of $\rho \circ \Phi_{2m}$ along D is $d\rho_F(d^2\Phi_{2m}[D, D]) = 2\sum_i H(D_i, D_i)$, which is nonnegative at a minimum: this is (2). If H or $-H$ were positive semidefinite, λ or $-\lambda$ would be nonnegative on Σ and vanish at the interior point F, so $\lambda = 0$; this proves (3). A common complex zero of the h_i is a singular point of F, excluded by $F \notin Z$: this is (4). □

Corollary 4. *In the situation of Proposition 3, for every representation* $F = \sum_{i \leq 2m} h_i^2$ *the* h_i *are linearly independent,* F *is not a sum of fewer than* $2m$ *squares,* $r := \operatorname{rank} H \leq m$*, and* $\operatorname{im} H_{\mathbb{C}}$ *contains no nonzero simple tensor* $x \otimes y$.

Proof. If the h_i were dependent, an orthogonal change of the $2m$ summands (which does not change F) would make $h_{2m} = 0$; then $D = (0, \ldots, 0, v)$ satisfies $\sum h_i D_i = 0$ for every v, and Proposition 3(2) gives $H(v, v) \geq 0$ for all v, against (3). A representation with fewer squares, completed with zeros, would be dependent. So $\dim W_h = 2m$ and $\operatorname{im} H \subseteq W_h^{\perp}$ has dimension at most $3m - 2m = m$. Finally $\operatorname{im} H_{\mathbb{C}} \subseteq (W_h)_{\mathbb{C}}^{\perp}$, and $x \otimes y \in (W_h)_{\mathbb{C}}^{\perp}$ means $h_i(x, y) = 0$ for all i, excluded by (4). □

4. The support of a multiplier

For a symmetric H with (1) let $A'' = \{x \in A : \ H(x \otimes b, v) = 0 \text{ for all } b \in B, \ v \in V\}$ and $a = \dim A - \dim A''$: the support of H in the first factor has dimension a, and H induces on $A' \otimes B$, for any complement A' of A'', a form of the same rank with trivial such radical.

Lemma 5. *Let* $H \neq 0$ *be symmetric with (1) and support of dimension* a *in the first factor. If* $\operatorname{im} H_{\mathbb{C}}$ *contains no nonzero simple tensor, then* $\operatorname{rank} H \geq a + 1$.

Proof. Restrict to $A' \otimes B$, complexify and write $H(u, v) = J(Tu, Tv)$ with $T \colon A'_{\mathbb{C}} \otimes B_{\mathbb{C}} \to E$ onto, $\dim E = r = \operatorname{rank} H$, and J a nondegenerate symmetric bilinear form on E. Put

$T_b x = T(x \otimes b)$. Trivial radical means $\bigcap_b \ker T_b = 0$, and (1) reads

$$J(T_b x, T_c v) = J(T_c x, T_b v). \tag{3}$$

A line gives a simple tensor. If $x \neq 0$ and $T_b x = \beta(b)e$ for a linear form $\beta \neq 0$ and $e \neq 0$, choose c with $\beta(c) = 1$; by (3), $J(e, T_b v) = \beta(b)\, J(e, T_c v)$ for all $v \in A'$, $b \in B$. The left side, as a linear form in $v \otimes b$, lies in $\operatorname{im} H_{\mathbb{C}}$ (T is onto, J nondegenerate) and is nonzero; the right side is the simple tensor $\beta \otimes J(e, T_c \cdot)$. So it suffices to find such an x when $r \leq a$.

Assume $r \leq a$. Choose b with $s = \operatorname{rank} T_b$ maximal, $U = \operatorname{im} T_b$, $N_0 = \ker T_b$. For every c,

$$T_c(N_0) \subseteq U \cap U^{\perp_J}. \tag{4}$$

Orthogonality follows from (3) with $x \in N_0$. For $T_c(N_0) \subseteq U$, write T_b as an identity block of size s and zeros; the coefficient of t in an $(s+1)$-minor of $T_b + tT_c$ bordering that block is an entry of $T_c \colon N_0 \to E/U$, and maximality of s forces it to vanish.

If $s = r < a$, then $U = E$, $U \cap U^{\perp_J} = 0$, and (4) gives a common kernel $N_0 \neq 0$, a contradiction. If $s = r = a$, T_b is invertible; complete b to a basis b, c, d. The operators $T_b^{-1}T_c$, $T_b^{-1}T_d$ are self-adjoint for $J(T_b \cdot, T_b \cdot)$ by (3), and (3) for the pair c, d makes their product self-adjoint as well, so they commute; a common eigenvector $x \neq 0$ gives $T_c x, T_d x \in \mathbb{C}\, T_b x$, a line. If $s < r$, put $R = U \cap U^{\perp_J}$; then $\dim N_0 = a - s \geq r - s \geq \dim R$. If $R = 0$ there is again a common kernel. Otherwise $T_c|_{N_0}, T_d|_{N_0} \colon N_0 \to R$ form a pencil: if $\dim N_0 > \dim R$ every member has a kernel, and if the dimensions agree the binary form $\det(\sigma T_c + \tau T_d)|_{N_0}$ is zero or of positive degree, so it vanishes at some $[\sigma : \tau] \in \mathbb{P}^1(\mathbb{C})$. Either way there is $x \in N_0 \setminus 0$ with $T_b x = 0$ and $T_c x, T_d x$ dependent, not both zero; they span a line. In all cases the first paragraph gives a contradiction. □

5. Induction and second variation

Proof of the upper bound in Theorem 1. We prove $\Sigma_{2m} = \Sigma$ by strong induction on $m \geq 2$. Suppose not, and take F, H as in Proposition 3; fix a representation $F = \sum_{i \leq 2m} h_i^2$ and $W = W_h$. By Corollary 4 and Lemma 5, $a + 1 \leq r \leq m$, so $1 \leq a \leq m - 1$. Let $A = A' \oplus A''$ orthogonally, $e = m - a$.

Case $e \geq 2$. Write $w = w' + w''$ with $w' \in A' \otimes B$, $w'' \in A'' \otimes B$. Since $A'' \otimes B$ lies in the radical of H and $W \subseteq \ker H$, the projection W' of W to $A' \otimes B$ lies in the kernel of $H|_{A' \otimes B}$, which has dimension $3a - r$. An orthogonal change of the $2m$ summands moves all the components in $A' \otimes B$ into at most $s = \dim W' \leq 3a - r$ of them; the other $2m - s$ summands are bilinear forms in the e variables of A'', and their sum of squares is a sum of squares in $e \times 3$ variables. As $2 \leq e < m$, by induction it is a sum of $2e$ squares. Hence F is a sum of at most $3a - r + 2(m - a) = 2m + a - r \leq 2m - 1$ squares, against Corollary 4.

Case $e = 1$. Then $a = m - 1$ and $r = m = a + 1$, so $W = \ker H$ and $A'' \otimes B \subseteq W$; let x_0 span A''. The coefficient map $\mathbb{R}^{2m} \to W$ of the representation is an isomorphism (Corollary 4); choosing an orthonormal basis of the preimage of $x_0 \otimes B$ and completing it, an orthogonal change of the summands gives $h_i = x_0 \otimes y_i$ $(i = 1, 2, 3)$ for a basis y_1, y_2, y_3 of B. Write H_{jl} for the blocks of $H|_{A' \otimes B}$ in this basis (symmetric, with $H_{jl} = H_{lj}$ by (1)) and $S = H_{11} + H_{22} + H_{33}$. For $v_1, v_2, v_3 \in A'$ let

$$D_i = \sum_{j,p} \varepsilon_{ijp}\, v_p \otimes y_j \quad (i = 1, 2, 3), \qquad D_i = 0 \quad (i > 3),$$

with ε the alternating symbol. Then $\sum_i h_i D_i = \sum_{i,j,p} \varepsilon_{ijp}(x_0 \otimes y_i)(v_p \otimes y_j) = 0$ by antisymmetry in i, j, and $\sum_i \varepsilon_{ijp}\varepsilon_{ilq} = \delta_{jl}\delta_{pq} - \delta_{jq}\delta_{pl}$ gives

$$\sum_i H(D_i, D_i) = \sum_p v_p^\top S v_p - \sum_{p,q} v_p^\top H_{pq} v_q.$$

By Proposition 3(2) this is nonnegative for all v, that is $L := I_3 \otimes S - H|_{A' \otimes B} \succeq 0$ in the block order. The three diagonal blocks of L sum to $3S - S = 2S$, so $S \succeq 0$. If $Sx = 0$, the three nonnegative numbers $L(x \otimes y_j, x \otimes y_j)$ sum to $2x^\top S x = 0$, so $L(x \otimes y_j, \cdot) = 0$ and hence $H(x \otimes y_j, \cdot) = 0$ for all j; by the choice of A', $x = 0$. So $S \succ 0$. The congruence by $\operatorname{diag}(S^{-1/2}, S^{-1/2}, S^{-1/2})$ turns $H|_{A' \otimes B}$ into $\widetilde{H}$ with $\widetilde{H} \preceq I_{3a}$, $\operatorname{tr} \widetilde{H} = \operatorname{tr}(S^{-1/2} S S^{-1/2}) = a$, $\operatorname{rank} \widetilde{H} = a + 1$ and the inertia of H, which by Proposition 3(3) has a negative eigenvalue. Then $\widetilde{H}$ has at most a positive eigenvalues, each at most 1, and a negative one, so $\operatorname{tr} \widetilde{H} < a$: a contradiction.

For $m = 2$ only $e = 1$ occurs, so the induction starts without hypothesis. □

6. The lower bound

Lemma 6. *The form $F_m = \sum_{i=1}^m x_i^2\big((y_2 - iy_1)^2 + y_3^2\big)$ is a sum of $2m$ squares of bilinear forms and of no fewer.*

Proof. The displayed expression has $2m$ squares. Write any representation as

$$F_m = \Big\| \sum_i x_i M_i y \Big\|^2, \qquad M_i \in \mathbb{R}^{s \times 3}.$$

The coefficient of x_i^2 gives $M_i^\top M_i$ of rank two with kernel spanned by $b_i = (1, i, 0)$. For $i \neq j$ the coefficient of $x_i x_j$ vanishes, so $M_i^\top M_j$ is antisymmetric; it kills b_j on the right and b_i on the left, hence, being antisymmetric, b_i and b_j on the right. An antisymmetric 3×3 matrix with two independent vectors in its kernel is zero. So the images of the M_i are m planes, pairwise orthogonal in $\mathbb{R}^s$, and $s \geq 2m$. □

Together with $\operatorname{qp}(\mathbb{P}^{m-1} \times \mathbb{P}^2) = m$ and [4, Theorem 1.3] this gives $\operatorname{BSR}(m, 3) \geq 2m$ twice, and with §5 it completes the proof of Theorem 1. □

7. Checks and formal status

The computational steps were checked with exact or modular arithmetic and controls that must fail: the regular point of Lemma 2 ($d\Phi_{2m}$ onto for $2 \leq m \leq 7$; with $Q = P$ the rank drops), the identity of §5 for random Hankel multipliers in a non-orthonormal basis, a numerical search in which every sampled $m \times 3$ form with $m = 4, 5, 6$ is a sum of $2m$ squares while F_m is not a sum of $2m - 1$, and an example showing that a multiplier of rank m without simple tensors in its image occurs for $m = 5$, so the case $e = 1$ is not empty. The scripts accompany this note, and on the companion site the rank at the regular point and the $2m$ squares of F_m are computed in the reader's browser. Part of the argument is checked in Isabelle/HOL, without `sorry`, in the public repository of formalizations: the lower bound $\operatorname{BSR}(m, 3) \geq 2m$ of Lemma 6 (every representation of F_m has at least $2m$ squares), the algebraic core of Lemma 2 (the three block symmetries force $M = 0$ for P cyclic and $Q = DP^{-1}$), and the case $e = 1$ of §5 (the second variation identity and the trace bound,

stated with rank at most $a+1$). Lemma 5, the case $e \geq 2$, the passage from the Hankel multiplier to the block symmetries, Proposition 3 and the assembly of Theorem 1 are not formalized.

Questions

Does the method give $\mathrm{BSR}(m,4)$? Our earlier note gives $\lfloor 8m/3 \rfloor \leq \mathrm{BSR}(m,4) \leq 3m+1$. In width four the multiplier of a boundary point has rank at most $4m - \mathrm{BSR}$, and the second variation would have to come from a structure replacing the alternating symbol on $\mathbb{R}^3$.

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